

DIMITRIS OIKONOMOU

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SUMMARY

Final-year PhD student in optimization for machine learning, with five first-author papers at ICLR, ICML and NeurIPS. My current research focuses on adaptive step-size methods and sharpness-aware minimization, with applications to deep learning and LLMs.

EDUCATION

Johns Hopkins University <i>PhD in Computer Science (Advisor: Dr. Nicolas Loizou)</i>	Baltimore, USA 08/2023 – 05/2027 (expected)
National Technical University of Athens <i>MSc in Data Science and Machine Learning</i> Thesis: Efficient Learning Algorithms of Time Varying Ranking Distributions, Advisor: Dr. Dimitris Fotakis	Athens, Greece GPA: 9.8/10
University of Göttingen <i>MSc in Mathematics</i> Thesis: Convergence Results for Entropic Transfer Operators, Advisor: Dr. Bernhard Schmitzer	Göttingen, Germany GPA: 1.2 (German scale; $\approx$ 9.7/10)
National and Kapodistrian University of Athens <i>BSc in Mathematics</i>	Athens, Greece GPA: 9.6/10

PUBLICATIONS

- [7] **Dimitris Oikonomou** and Nicolas Loizou. *Curvature-Scaled Sharpness-Aware Minimization: Convergence Beyond Smoothness*. Under Submission.
- [6] **Dimitris Oikonomou**, Matthew Buchholz, Yuen-Man Pun, Robert M. Gower, Nicolas Loizou. *Taking the Road Less Scheduled with Adaptive Polyak Steps*. In **NeurIPS 2026**.
- [5] **Dimitris Oikonomou** and Nicolas Loizou. *Adaptive Sharpness-Aware Minimization with a Polyak-type Step size: A Theory-Grounded Scheduler*. In **ICML 2026**. [code]
- [4] **Dimitris Oikonomou** and Nicolas Loizou. *Safeguarded Stochastic Polyak Step Sizes for Non-smooth Optimization: Robust Performance Without Small (Sub)Gradients*. In **ICML 2026**. [code]
- [3] Robert M. Gower, Guillaume Garrigos, Nicolas Loizou, **Dimitris Oikonomou**, Konstantin Mishchenko, Fabian Schaipp. *Analysis of an Idealized Stochastic Polyak Method and its Application to Black-Box Model Distillation*. In Transactions on Machine Learning Research (**TMLR**), 2026.
- [2] **Dimitris Oikonomou** and Nicolas Loizou. *Sharpness-Aware Minimization: General Analysis and Improved Rates*. In **ICLR 2025**. [code]
- [1] **Dimitris Oikonomou** and Nicolas Loizou. *Stochastic Polyak Step-sizes and Momentum: Convergence Guarantees and Practical Performance*. In **ICLR 2025**. [code]

TALKS & PRESENTATIONS

Poster [6], Conference on Neural Information Processing Systems (NeurIPS 2026)	Atlanta, Dec 2026 (upcoming)
Posters [4, 5], International Conference on Machine Learning (ICML 2026)	Seoul, Jul 2026
Invited Talk, SIAM Conference on Optimization (OP26)	Edinburgh, Jun 2026
Invited Talk, ELLIIT Focus Period on Optimization for Learning	Lund, May 2026
Poster, ELLIIT Symposium on Optimization for Learning	Lund, May 2026
Posters [1, 2], Greeks in AI 2025	Athens, Jul 2025
Posters [1, 2], International Conference on Learning Representations (ICLR 2025)	Singapore, Apr 2025

HONORS & AWARDS

Mathematical Olympiads International Mathematical Olympiad (IMO): <i>Bronze Medal</i> Greek Mathematical Olympiad: <i>Silver Medal</i> Greek Mathematical Olympiad: <i>Bronze Medal</i>	Santa Marta, Colombia, 2013 Athens, Greece, 2014 Athens, Greece, 2013
Awards & Scholarships Gold Reviewer Award , International Conference on Machine Learning (ICML 2026) JHU MINDS Data Science Fellowship DAAD Scholarship (German Academic Exchange Service)	2026 2024, 2025 2019–2020

TEACHING & PROFESSIONAL SERVICE

Head Teaching Assistant: <i>Computer Vision</i>	JHU, Fall 2024
Mentor: JHU MINDS summer outreach program (high-school student)	JHU, Summer 2024
Reviewer: ICML, NeurIPS, ICLR, AISTATS; TMLR, IEEE Trans. Signal Processing, Information and Inference	